THE EULER CHARACTERISTIC OF THE TWISTS OF A COHERENT SHEAF ON PROJECTIVE SPACE IS A POLYNOMIAL IN THE TWIST

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites five cross-chapter labels (including a remark), each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The author's closing sentence omits the proof of the elementary arithmetic fact that a function with polynomial first difference is a polynomial, pointing to the exact algebra lemma; that sentence is kept verbatim and the omission is a routine verification.

Allowed standard prerequisites (editorial).

- The hyperplane section sequence and its cohomology consequence. Stacks Project, Varieties, Lemma lemma-exact-sequence-induction (tag 0ABH) and Remark remark-exact-sequence-induction-cohomology (tag 0ABI). Supplies the difference identity between consecutive Euler characteristics.
- Numerical polynomials: a function with polynomial difference is polynomial. Stacks Project, Algebra, Lemma algebra-lemma-numerical-polynomial (tag 0BWC). The arithmetic fact invoked at the end of the proof.

Source reference. Original lemma 08AB: sources/stacks-chow/tags.txt, source lines 10526–10526. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08A0: sources/stacks-chow/tags.txt, source lines 10515–10515. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08AA: sources/stacks-chow/tags.txt, source lines 10525–10525. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0EGK: sources/stacks-chow/tags.txt, source lines 18100–18100. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00JZ: sources/stacks-chow/tags.txt, source lines 733–733. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let $n \geq 0$. Let $\mathcal{F}$ be a coherent sheaf on $\mathbf{P}^n_k$. The function*

$$d \longmapsto \chi(\mathbf{P}^n_k, \mathcal{F}(d))$$

is a polynomial.

Proof. We prove this by induction on n. If $n = 0$, then $\mathbf{P}^n_k = \mathrm{Spec}(k)$ and $\mathcal{F}(d) = \mathcal{F}$. Hence in this case the function is constant, i.e., a polynomial of degree 0. Assume $n > 0$. By Lemma 08AB we may assume k is infinite. Apply Lemma 08A0. Applying Lemma 08AA to the twisted sequences $0 \to \mathcal{F}(d-1) \to \mathcal{F}(d) \to i_*\mathcal{G}(d) \to 0$ we obtain

$$\chi(\mathbf{P}^n_k, \mathcal{F}(d)) - \chi(\mathbf{P}^n_k, \mathcal{F}(d-1)) = \chi(H, \mathcal{G}(d))$$

See Remark 0EGK. Since $H \cong \mathbf{P}^{n-1}_k$ by induction the right hand side is a polynomial. The lemma is finished by noting that any function $f : \mathbf{Z} \to \mathbf{Z}$ with the property that the map $d \mapsto f(d) - f(d-1)$ is a polynomial, is itself a polynomial. We omit the proof of this fact (hint: compare with Algebra, Lemma 00JZ). □